

Embedded System and Internet of Things

Task:-01

A) Introduction:-

1.What is an embedded system?

- An embedded system is a special-purpose computer system designed to perform a specific function within a larger system.
- It consists of hardware (microcontroller/microprocessor) and software (firmware).
- Designed for real-time operation, low power, and high reliability.
- Examples: washing machine controller, microwave oven, digital camera.

2. Real Time applications of Embedded System:-

- **Smart Home:** Smart lights, smart thermostats, security systems, voice assistants
- **Automotive:** Engine control unit (ECU), ABS, airbag system, infotainment systems
- **Healthcare:** Pacemakers, ECG machines, patient monitoring systems, infusion pumps
- **Robotics:** Industrial robots, robotic arms, autonomous robots
- **Consumer Electronics:** Smartphones, smart TVs, cameras
- **Industrial Automation:** PLCs, CNC machines, sensors and actuators

3. Difference Between Microprocessor and Microcontroller:-

Microcontroller	Microprocessor
Single-chip system	CPU only
Contains CPU, RAM, ROM, I/O ports	Requires external RAM, ROM, I/O
Low power consumption	High power consumption
Used in embedded systems	Used in general-purpose computers
Low cost	Expensive
Simple architecture	Complex architecture
Dedicated to specific tasks	Handles multiple tasks
Examples: Arduino (ATmega), PIC, 8051	Examples: Intel i5/i7, AMD Ryzen

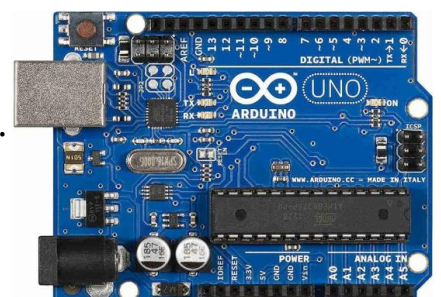
B) Component Study:-

1. Role of Microcontroller in Embedded Systems:-

- A microcontroller (MCU) is a compact integrated circuit designed to control specific operations in embedded systems.
- It includes CPU, memory (RAM/Flash), and I/O peripherals on a single chip.
- Used for real-time control, data acquisition, and automation.

1. Arduino:-

- Open-source **microcontroller platform** (based on ATmega328, ATmega2560, etc.).
- Easy to program using **Arduino IDE (C/C++)**.
- Ideal for **beginners, students, and prototyping**.
- Used in **IoT projects, automation, robotics**.



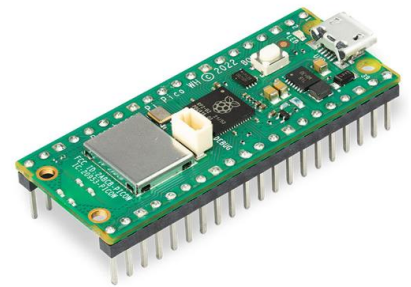
2. ESP32:-

- High-performance **microcontroller with built-in Wi-Fi and Bluetooth.**
- Dual-core processor with **low power modes.**
- Supports **IoT and cloud-based applications.**
- Used in **smart home systems, wireless sensors, wearables.**



3. Raspberry Pi Pico:-

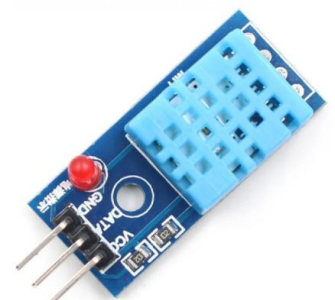
- Based on **RP2040 microcontroller.**
- Dual-core ARM Cortex-M0+ processor.
- Supports **C/C++ and MicroPython.**
- Used in **real-time control, automation, and learning embedded systems.**



2. Basic Sensors Used in Embedded Systems:-

1. Temperature Sensor:-

- Measures **ambient temperature.**
- Converts temperature into an electrical signal.
- Examples: **LM35, DHT11, DS18B20.**
- Applications:
 - Weather monitoring
 - HVAC systems
 - Medical devices



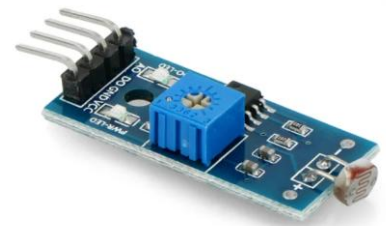
2. Motion Sensor:-

- Detects **movement of objects or humans**.
- Common type: **PIR (Passive Infrared) sensor**.
- Applications:
 - Security systems
 - Automatic lighting
 - Smart surveillance



3. Light Sensor:-

- Detects **light intensity**.
- Common types: **LDR (Light Dependent Resistor)**, photodiode.
- Applications:
 - Automatic street lights
 - Brightness control in displays
 - Smart home lighting

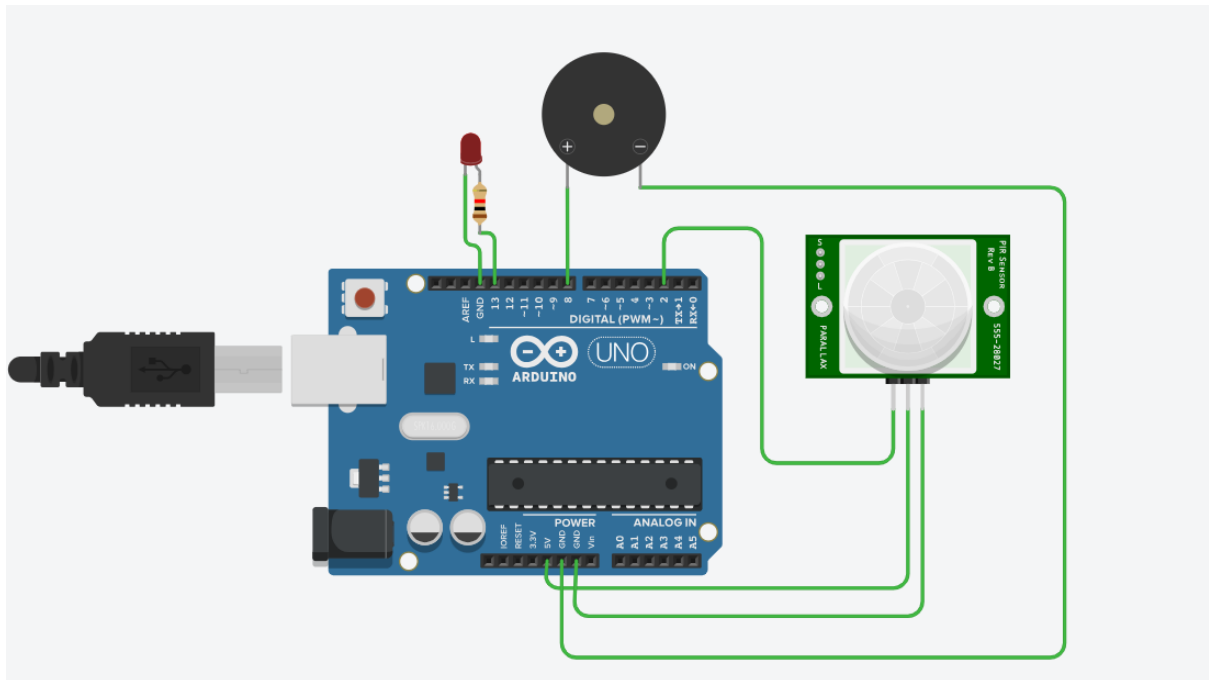


C) Mini Project:- Motion Detection Alaram

1. Working Principle:-

- The **PIR sensor** detects changes in **infrared radiation** caused by human or object movement.
- When motion is detected, the PIR sensor outputs a **HIGH signal**.
- The microcontroller reads this signal.
- If motion is detected:
 - **Buzzer sounds** and/or **LED turns ON**
- If no motion is detected:
 - Buzzer/LED remains **OFF**

2. Circuit Diagram:-



3. Source Code:-

```
// Pin definitions

int pirPin = 2;    // PIR sensor output pin

int ledPin = 13;   // LED pin

int buzzerPin = 8; // Buzzer pin

void setup() {

  pinMode(pirPin, INPUT);

  pinMode(ledPin, OUTPUT);

  pinMode(buzzerPin, OUTPUT);

  Serial.begin(9600);
```

```
}

void loop() {

    int motion = digitalRead(pirPin);

    if (motion == HIGH) {

        digitalWrite(ledPin, HIGH);

        digitalWrite(buzzerPin, HIGH);

        Serial.println("Motion Detected!");

        delay(1000);

    } else {

        digitalWrite(ledPin, LOW);

        digitalWrite(buzzerPin, LOW);

        Serial.println("No Motion");

        delay(500);

    }

}
```

4. Real World Applications:-

- Home security systems
- Office and industrial security
- Automatic door control
- Smart surveillance systems